Practice Questions

1. Write a Python program to create a list of 10 integers entered by the user and then display the list in reverse order without using the reverse() method or slicing [:: -1].
2. Write a Python program that takes a list of numbers as input and prints a new list containing only the even numbers from the original list.
3. Write a Python program to find and print the largest and smallest element in a list without using built-in max() and min() functions.
4. Write a Python program that accepts a list of strings and prints the list after removing all duplicate entries (preserve the original order of first occurrence).
5. Write a Python program that takes 8 numbers as input from the user, stores them in a tuple, and then prints the sum, maximum, minimum, and average of the numbers.
6. Write a Python program to create a tuple containing names of 5 students. Then accept a name from the user and check if it exists in the tuple. Print appropriate messages.
7. Write a Python program that creates two tuples and prints a new tuple containing elements common to both tuples (without duplicates).
8. Write a Python program to accept 6 integers from the user and store them in a set. Then remove all even numbers from the set and print the remaining set.
9. Write a Python program that takes two sets of integers as input from the user and prints Union, Intersection, Difference (first – second) and Symmetric difference
10. Write a Python program to create a set of 10 random numbers between 1 and 20, then print whether the set contains the number 15 or not.
11. Write a Python program that accepts a sentence from the user and creates a set of all unique vowels (a, e, i, o, u) present in the sentence (case-insensitive).
12. Write a Python program to create a dictionary of 5 students where key is the student's name and value is their mark. Then print the names of students who scored more than 75 marks.
13. Write a Python program that takes a dictionary of items and their prices, then calculates and prints the total bill amount.
    Example input: `{'Apple': 50, 'Banana': 20, 'Milk': 40}` → Total = 110
14. Write a Python program to count the frequency of each character in a given string and store the result in a dictionary.
    Example: "hello" → `{'h':1, 'e':1, 'l':2, 'o':1}`
15. Write a Python program to create a dictionary where keys are numbers from 1 to 10 and values are squares of the keys. Then print the dictionary in sorted order of keys.
16. Write a Python program that accepts roll number and name of students and stores them in a dictionary. Keep accepting until the user enters "stop" as roll number. Finally print the entire phonebook-like dictionary.
17. Write a Python program to convert a list of tuples `[('A', 65), ('B', 66), ('C', 67)]` into a dictionary and then print the dictionary.